

NAPCORE Helpdesk: Trusted FAQ + RAG Architecture

Presentation Draft

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Problem And Goal

- ▶ Need a trustworthy helpdesk for multimodal standards implementation.
- ▶ Main risk is hallucinated or uncited AI guidance.
- ▶ Goal is grounded answers with traceable references.

System Context (C4 Level 1)

Five stakeholder groups share one helpdesk interface:

- ▶ PTO (Public Transport Operator) — responsible for daily service delivery
- ▶ PTA (Public Transport Authority) — sets policy and compliance standards
- ▶ Developer — implements integrations and APIs
- ▶ ITS System Integrator — connects helpdesk to broader mobility ecosystem
- ▶ Ticketing Agent — manages passenger-facing systems

External systems:

- ▶ Approved GitHub repositories (NeTEx-CEN/NeTEx) — knowledge source
- ▶ LLM provider — grounded generation only; no unconstrained model memory

Key architectural constraint:

- ▶ All answers sourced from approved repositories only

Scope

- ▶ Standards: Transmodel, NeTEx, SIRI, OJP/OpRa, DATEX II.
- ▶ Priority source: NeTEx-CEN/NeTEx and approved companion repositories.
- ▶ User channel: GUI-based technical Q and A.

Core Approach

- ▶ FAQ-first with RAG fallback.¹
- ▶ FAQ-first: return approved canonical answers when confidence is high.
- ▶ RAG fallback: retrieve source chunks then generate grounded answer.
- ▶ If evidence is insufficient, abstain explicitly.

¹RAG fallback means that when no high-confidence FAQ answer is available, the system retrieves relevant source chunks and then generates an answer grounded in that retrieved evidence.

Trust And Safety

- ▶ Repository allowlist enforces knowledge boundary.
- ▶ Generation limited to retrieved evidence.
- ▶ Citation gate blocks uncited publication.
- ▶ Editorial gate enforces draft, review, approve, publish states.

Functional Requirements Highlights

- ▶ FR-001: developer question answering in GUI.
- ▶ FR-004: ingestion and indexing of approved GitHub sources.
- ▶ FR-005: retrieval and grounded generation.
- ▶ FR-006: anti-hallucination and boundary enforcement.
- ▶ FR-002 and FR-003: editorial curation and FAQ promotion.

Data Model Logic

- ▶ Source governance: approved_repositories -> source_documents -> source_chunks.
- ▶ FAQ lifecycle: faq_entries -> faq_versions -> review_events.
- ▶ Runtime telemetry: question_events -> retrieval_events.
- ▶ Evidence mapping: answer_evidence_links ties answers to source chunks.

Why Hallucinations Are Prevented

- ▶ Only approved repositories are retrievable.
- ▶ Unsupported claims fail citation and support checks.
- ▶ Evidence links provide claim-to-source traceability.
- ▶ Unknown questions produce abstention, not speculation.

User Experience Flow

- ▶ User submits question in GUI.
- ▶ System attempts FAQ match first.
- ▶ On miss or low confidence, system executes RAG retrieval plus generation.
- ▶ Response includes references and confidence signal.
- ▶ Feedback is captured for continuous improvement.

Operations And Governance

- ▶ Reviewer decisions are logged and auditable.
- ▶ Editorial operations run through Django Admin.
- ▶ Recurring intents are promoted into curated FAQ backlog.
- ▶ Re-indexing handles source and standards updates.
- ▶ Metrics monitor quality and safety continuously.

MVP Success Metrics

- ▶ Citation coverage for published FAQs: 100%.
- ▶ Unsupported-claim publication rate: 0%.
- ▶ P95 latency: ≤ 8 seconds.
- ▶ Retrieval success on covered intents: $\geq 90\%$.

Current Artifacts

- ▶ Functional requirements document.
- ▶ RAG architecture and database logic documents.
- ▶ Technology baseline: Django 5 + DRF + Django Admin + Celery/Redis.
- ▶ PlantUML database model and SQL migration.
- ▶ Diagram render target for reproducible outputs.

Next Steps

1. Finalize repository allowlist and ingestion profile.
2. Implement Django/DRF orchestration API for FAQ-first plus RAG fallback.
3. Configure Django Admin editorial workflow and publication gates.
4. Run pilot validation against KPI targets.

Appendix: Example Questions

- ▶ How to use NeTEx for exchanging a timetable?
- ▶ How to implement IDs for a Stop Place registry?

Expected outputs: - practical implementation guidance, - explicit citations, - abstention where evidence is insufficient.